

Name:Ravi Shankar Tripathi

Github Repo:https://github.com/RAshtr/Flask_and_MongoDB_Assignment

Task 1 Task 1: JSON API Route

Building Flask App & Cloud Database Connection:

Objective:Create a Flask web server and connect it to MongoDB Atlas using PyMongo to handle form inputs and store submissions securely in the cloud.

Source Code and output Screenshots:

[app.py](#):

```
import os
from dotenv import load_dotenv
from flask import Flask, render_template, request
from pymongo import MongoClient

load_dotenv()

app = Flask(__name__)

MONGO_URI = os.getenv("MONGO_URI")
# Add tlsAllowInvalidCertificates to bypass Windows OpenSSL handshake mismatch
client = MongoClient(MONGO_URI, tlsAllowInvalidCertificates=True)
db = client.get_database("student_assignment_db")
collection = db.user_submissions

@app.route("/")
def home():
    return render_template("form.html")

@app.route("/submit", methods=["POST"])
def submit():
    user_name = request.form.get("name")
    user_email = request.form.get("email")
    user_message = request.form.get("message")

    submission_data = {
        "name": user_name,
        "email": user_email,
        "message": user_message
    }
```

```

collection.insert_one(submission_data)

return render_template("success.html", name=user_name)

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=5000, debug=False)

```

templates/form.html :

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Contact Form</title>
</head>
<body>
    <h2>User Submission Form</h2>
    <form action="/submit" method="POST">
        <label>Name:</label><br>
        <input type="text" name="name" required><br><br>

        <label>Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Message:</label><br>
        <textarea name="message" required></textarea><br><br>

        <button type="submit">Submit</button>
    </form>
</body>
</html>

```

Task 1 Explanation:

Loaded the MongoDB connection string safely from .env using python-dotenv.
 Initialized MongoClient with tlsAllowInvalidCertificates=True to prevent
 SSL handshake errors on Windows .

Created two routes: GET / to show the input form and POST /submit to save
 incoming form data into MongoDB Atlas collection user_submissions .



127.0.0.1:5000

User Submission Form

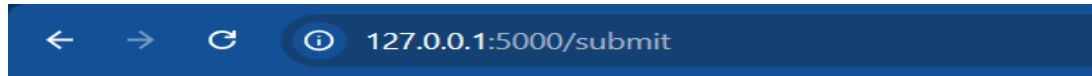
Name:



Please fill out this field.

Message:

Submit



Thank You, Ravi Tripathi!

Your details have been successfully saved to MongoDB Atlas.

[Go Back to Form](#)

Task 2: Frontend Form & Success UI :

Objective: Design an HTML input form with field validation and a dynamic success response page.

templates/success.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Submission Successful</title>
</head>
```

```

<body>
  <h2>Thank You, {{ name }}!</h2>
  <p>Your details have been successfully saved to MongoDB Atlas.</p>
  <a href="/">Go Back to Form</a>
</body>
</html>

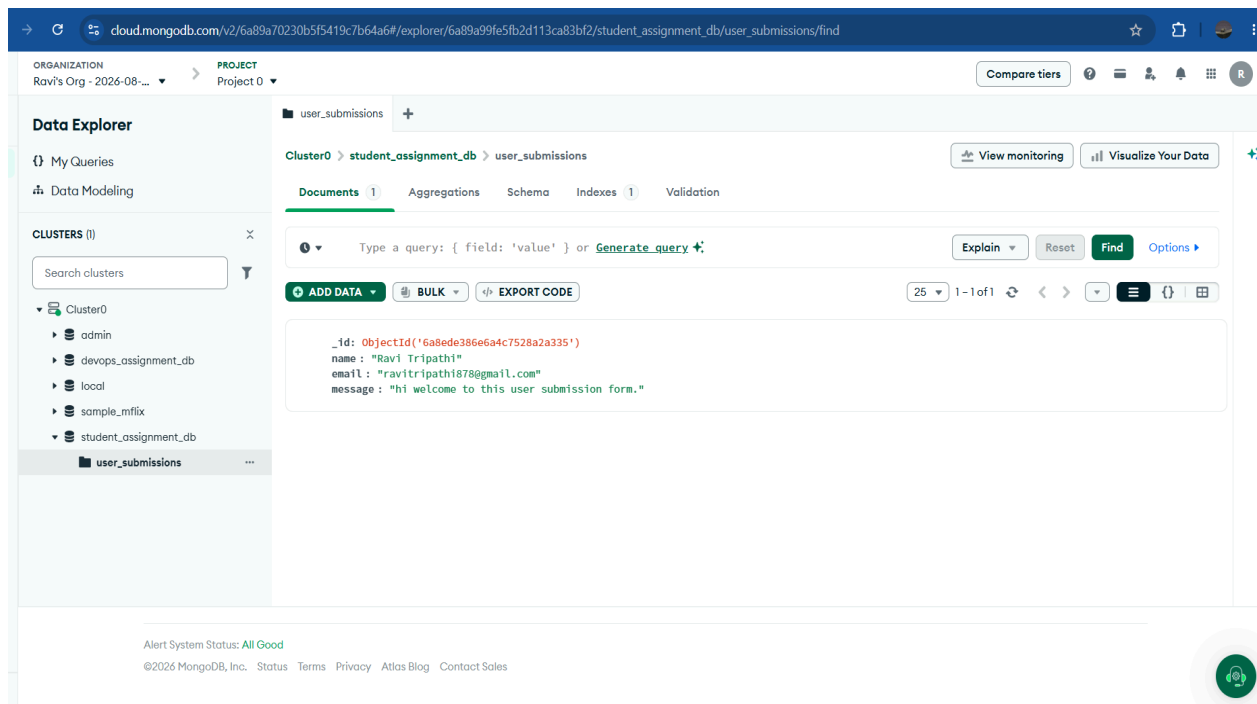
```

Task 2: Frontend Form with MongoDB Atlas:

Task 2 Explanation: Used standard HTML5 required attributes in form.html so the browser stops empty submissions directly on the frontend.

The form sends a POST request to /submit .

When MongoDB successfully stores the record, Flask renders success.html and passes the user's name dynamically to confirm data submission .



Containerization with Docker :

objective:Create a Dockerfile to package the application, dependencies, and environment into a standalone container image .

code:(dockerfile)

```
FROM python:3.10-slim
```

```
WORKDIR /app
```

```
COPY requirements.txt requirements.txt
```

```
RUN pip install --no-cache-dir -r requirements.txt
```

```
COPY . .
```

```
EXPOSE 5000
```

```
CMD ["python", "app.py"]
```

Explain:Used python:3.10-slim as the base image to keep build size small.

Copied dependencies and project files, exposed port 5000, and defined the entry point command.

Executed build and run commands in terminal.

Terminal command:

```
docker build -t flask-mongo-app .
```

```
docker run -d -p 5000:5000 --env-file .env --name flask_app_container flask-mongo-app
```

```
docker ps
```

Flask_and_MongoDB_Ravi_Shankar_Tripathi

app.py

```
20 @app.route("/submit", methods=["POST"])
21 def submit():
22     user_name = request.form.get("name")
23     user_email = request.form.get("email")
24     user_message = request.form.get("message")
25
26     submission_data = {
27         "name": user_name,
28         "email": user_email,
29         "message": user_message
30     }
31     collection.insert_one(submission_data)
32
33     return render_template("success.html", name=user_name)
```

PS C:\Users\sukhram prajapati\Documents\Flask_and_MongoDB_Ravi_Shankar_Tripathi> docker build -t flask-mongo-app .

```
-> sha256:418/9110350e9107600c00e11965910a5396d58485183a6741062c1074240 1.75kB / 1.75kB 0.0s
-> sha256:6601e925f15054c0bc6450eef1bee5c2562ff08909f4fa1ad0ca1e8a71834 5.48kB / 5.48kB 0.0s
-> sha256:6310eb1bf4251731feab01e8f632bf5e2d75a65ccad97f420b1f83ce457be 29.79MB / 29.79MB 101.7s
-> sha256:4f9c6a734391d1b92c16f2bc07462577523a9233c1685ba9ce4942857e91442d 13.91MB / 13.91MB 87.3s
-> sha256:5466a598b0f99c3e409980284227593fe59ca1cd23ad506ea5186efd6320b3c3 1.29MB / 1.29MB 50.9s
-> sha256:bbfc75a06bd204d0ffa19e93ccc8b0d95f06d4a469a030992b3d2b10429d2 249B / 249B 62.9s
-> extracting sha256:6310eb1bf4251731feab01e8f632bf5e2d75a65ccad97f420b1f83ce457be 13.0s
-> extracting sha256:5466a598b0f99c3e409980284227593fe59ca1cd23ad506ea5186efd6320b3c3 1.4s
-> extracting sha256:4f9c6a734391d1b92c16f2bc07462577523a9233c1685ba9ce4942857e91442d 9.4s
-> extracting sha256:bbfc75a06bd204d0ffa19e93ccc8b0d95f06d4a469a030992b3d2b10429d2 0.0s
-> [2/5] WORKDIR /app 6.2s
-> [3/5] COPY requirements.txt requirements.txt 0.6s
-> [4/5] RUN pip install --no-cache-dir -r requirements.txt 36.2s
-> [5/5] COPY . . 1.6s
-> exporting to image 2.0s
-> exporting layers 1.5s
-> writing image sha256:a6b7eb59bad729776598932b385346d2519573b34cab44f35f556a20d9e3a96 0.1s
-> naming to docker.io/library/flask-mongo-app 0.1s
```

PS C:\Users\sukhram prajapati\Documents\Flask_and_MongoDB_Ravi_Shankar_Tripathi

Ask about your code

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

app.py

Explore and understand your code

Models

Local

main* 0 0 0

Not Committed Yet Ln 31, Col 43 Spaces: 4 UTF-8 CRLF Python Python 3.12 (64-bit) Prettier

Search

Thunderstorm w... In effect

ENG IN

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